Brandon Viaje

647-721-4158 | viajebrandon@gmail.com | linkedin.com/in/brandonviaje | github.com/brandonviaje

Education

Ontario Tech University — Oshawa, ON
*Bachelor of Science in Computer Science Co-op (GPA: **4.03/4.30**)* — *Apr 2027*

- Honors: President's List (6x)
- Relevant Coursework: Compilers, Data Structures & Algorithms, Operating Systems, Machine Learning

Projects

Web Crawler | C++, MongoDB, libcurl, Python

- Archived **50,000+ web pages** from a discovery frontier of **4 million+ unique URLs** by engineering a **multithreaded web crawler** to handle high-throughput HTTP requests
- Achieved **<10ms search latency** by integrating MongoDB Atlas Search to index scraped data, enabling full-text keyword queries and relevance scoring
- Eliminated race conditions and handled high-volume I/O by implementing **thread-safe queues** and **synchronization primitives**

GEMM Accelerator | C++, CUDA, cuBLASLt, CMake

- Decreased GEMM latency by **82%** compared to an FP32 baseline by engineering a custom **INT8 quantization** data pipeline
- Achieved **46.84 TFLOPS** of GPU throughput by writing **custom CUDA kernels** and utilizing warp-level primitives to reduce memory bandwidth bottlenecks
- Maintained strict numerical stability (**MSE < 2.0**) by querying **cuBLASLt** heuristic engines and designing a custom dequantization kernel

Chess Engine | C, Google Cloud, Lichess API

- Achieved **1.2M positions/sec** evaluation speed on a single thread by engineering a UCI chess engine in C using **Bitboards**
- Increased search depth and efficiency by implementing **Alpha-Beta Pruning, Negamax algorithms**, and **transposition tables** for caching
- Facilitated **100+ automated games** with zero downtime by deploying a **24/7 matchmaking bot** to **Google Cloud** via the Lichess API

ML Inference Engine | C++, ONNX, Google Protobuf

- Reduced model initialization time and memory footprint by leveraging **Google Protobuf** to deserialize model weights into lightweight data structures
- Optimized **ONNX computation graphs** and ensured deterministic execution by implementing **topological sorting** to resolve node dependencies
- Built **scalable C++ infrastructure** for deploying machine learning models with reduced runtime overhead

Experience

Auto Body Assistant — Jul 2022 – Aug. 2023
Impera Auto Collision — *Toronto, ON*

- Assisted mechanics with daily vehicle repairs, component assembly, and maintaining an **organized workflow** during peak shop hours
- Managed **tool inventory** and ensured a safe, efficient workspace in a fast-paced, hands-on environment
- Demonstrated strong reliability, communication, and attention to detail while coordinating with team members to meet **daily repair targets**

Technical Skills

Languages: C++, C, Python, JavaScript, HTML/CSS, React
Tools & Infrastructure: CUDA, cuBLAS, Git, TensorFlow, PyTorch, libcurl, Linux/Unix
Databases: MongoDB, PostgreSQL, MySQL